
Coarsening Confers Robustness: Causal Bayesian Optimization under Directed Acyclic Graph Misspecification

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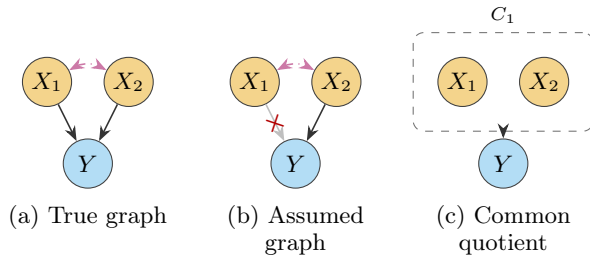


Figure 1: The PARALLEL PARENT construction. Deleting $X_1 \rightarrow Y$ changes the fine exploration set but not the quotient because the cluster edge $C_1 \rightarrow Y$ remains through $X_2 \rightarrow Y$. Orange nodes are manipulable, blue is the target, and the dash-dotted edge denotes latent confounding.

Abstract

1 Introduction

Causal Bayesian optimization (CBO) [Aglietti et al., 2020] searches for an intervention that optimizes a target while using observational data to reduce the number of costly experiments. Its assumed causal graph determines which intervention sets are considered and which causal effects can be used as prior means. A misspecified graph can consequently do more than slow learning: it can exclude the best intervention from the search space.

Figure 1 illustrates the failure mode. The confounded parents X_1 and X_2 jointly determine Y , and the optimum requires setting both variables. If the hypothesis graph deletes $X_1 \rightarrow Y$, the fine-graph optimizer discards every intervention containing X_1 . If the two variables are instead represented by the cluster $C_1 = \{X_1, X_2\}$, the edge $C_1 \rightarrow Y$ survives through $X_2 \rightarrow Y$ and the whole-cluster intervention remains available.

This observation motivates a deliberate change of resolution. QCBO receives a partition and its quotient graph as inputs and restricts interventions to unions of manipulable clusters. Such clusters may encode a known subsystem, a policy module, or another domain-defined unit whose coarse causal relationships are more reliable than its internal structure. The present work does not learn the partition or the quotient graph. Coarsening removes sensitivity to distinctions within clusters, but it can also hide useful fine-grained actions.

Our contributions are threefold. First, we formalize the *quotient contract*: every graph-dependent component of the optimizer factors through the quotient. This yields exact trajectory invariance over the class of fine graphs with the same quotient and a worst-case regret separation from fine-graph methods. Second, we characterize the price of restricting the action space to cluster unions, establish monotonicity under refinement, and give structural conditions for zero price. Third, we instantiate the contract for arm-based, dynamic, and model-based causal Bayesian optimizers and evaluate the predicted invariances and trade-offs. Proofs, complete dataset graphs, and additional results are provided in the appendices.

2 Problem setup and quotient construction

2.1 Causal Bayesian optimization

Let $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, F, P(\mathbf{U}) \rangle$ be a fixed structural causal model (SCM) with target $Y \in \mathbf{V}$ and manipulable variables $\mathbf{X} \subseteq \mathbf{V}$ [Pearl, 2009]. Its latent projection is an acyclic directed mixed graph (ADMG) $\mathcal{G}$ whose directed edges represent causal relations and whose bidirected edges represent latent confounding. Throughout, population expectations and intervention outcomes are defined by this fixed true SCM. A hypothesis graph supplied to an optimizer can change its decisions, but never the data-generating distribution.

An intervention arm is a nonempty set $A \subseteq \mathbf{X}$. Let $D(A)$ denote the Cartesian intervention domain for the variables in A . For minimization, the population value of an arm is

$$\mu^*(A) := \min_{a \in D(A)} \mathbb{E}_{\mathcal{M}}[Y \mid \text{do}(A = a)]. \quad (1)$$

CBO maintains a Gaussian process (GP) for each retained arm. Its exploration set consists of the graph's minimal intervention sets (MIS) [Lee and Bareinboim, 2018, Aglietti et al., 2020]. Writing $\mathcal{G}_{\bar{A}}$ for the graph obtained by deleting incoming edges to A , the MIS are

$$\mathbf{ES}(\mathcal{G}) = \text{MIS}(\mathcal{G}) := \left\{ A \subseteq \mathbf{X} : \begin{array}{l} A \neq \emptyset, \\ A \subseteq \text{An}(Y) \text{ in } \mathcal{G}_{\bar{A}} \end{array} \right\}. \quad (2)$$

For every retained arm A , CBO places a GP prior on the interventional response surface

$$f_A(a) := \mathbb{E}_{\mathcal{M}}[Y \mid \text{do}(A = a)], \quad f_A \sim \mathcal{GP}(m_A^H, k_A). \quad (3)$$

The prior mean is a plug-in estimate of the causal effect whenever that effect is identified under the hypothesis graph H . In particular, if a complete identification procedure returns the functional

$$p_H(y \mid \text{do}(A = a)) = \mathcal{J}_A^H(p)(y; a), \quad (4)$$

then, for an estimate $\hat{p}$ of the observational distribution,

$$m_A^H(a) = \int y \mathcal{J}_A^H(\hat{p})(y; a) dy. \quad (5)$$

For example, if Z is a valid adjustment set, this reduces to the familiar backdoor estimator

$$m_A^H(a) = \int \hat{\mathbb{E}}[Y \mid A = a, Z = z] d\hat{P}(z). \quad (6)$$

Under causal sufficiency, the truncated factorization provides the g -computation functional

$$\hat{p}(v \setminus a \mid \text{do}(a)) = \prod_{V_i \in \mathbf{V} \setminus A} \hat{p}(v_i \mid \text{pa}_i), \quad (7)$$

which is integrated against Y to obtain Equation (5). Equation (7) is one identified functional, not a substitute for complete identification in an ADMG; front-door and more general functionals may instead be required.

Let $K_{A,t}$ be the kernel matrix of the intervention values in $\mathcal{D}_{A,t}$, $k_{A,t}(a)$ the vector of covariances between those values and a , and $\mathbf{y}_{A,t}$ their observed outcomes. Define $C_{A,t} := K_{A,t} + \sigma_{A,\epsilon}^2 I$ for independent Gaussian observation noise of variance $\sigma_{A,\epsilon}^2$. The arm-wise GP posterior is

$$\begin{aligned} \mu_{A,t}(a) &= m_A^H(a) + k_{A,t}(a)^\top C_{A,t}^{-1}(\mathbf{y}_{A,t} - \mathbf{m}_{A,t}^H), \\ \sigma_{A,t}^2(a) &= k_A(a, a) - k_{A,t}(a)^\top C_{A,t}^{-1} k_{A,t}(a), \end{aligned} \quad (8)$$

where $\mathbf{m}_{A,t}^H$ evaluates the causal prior mean at the observed intervention values. For minimization, expected improvement over the incumbent y_t^{best} is

$$\begin{aligned} z_{A,t}(a) &= \frac{y_t^{\text{best}} - \mu_{A,t}(a)}{\sigma_{A,t}(a)}, \\ \text{EI}_{A,t}(a) &= (y_t^{\text{best}} - \mu_{A,t}(a)) \Phi(z_{A,t}(a)) \\ &\quad + \sigma_{A,t}(a) \phi(z_{A,t}(a)), \end{aligned} \quad (9)$$

with the deterministic limit used when $\sigma_{A,t}(a) = 0$. CBO selects the arm and value maximizing $\text{EI}_{A,t}(a)/c(A, a)$, where $c(A, a)$ is the intervention cost. A common alternative is GP-UCB [Srinivas et al., 2010]:

$$\begin{aligned} \text{UCB}_{A,t}(a) &= \mu_{A,t}(a) + \sqrt{\beta_t} \sigma_{A,t}(a), \\ \text{LCB}_{A,t}(a) &= \mu_{A,t}(a) - \sqrt{\beta_t} \sigma_{A,t}(a), \end{aligned} \quad (10)$$

where UCB is maximized and LCB is minimized. Our arm-based CBO and QCBO experiments use cost-normalized EI; the model-based experiments retain the base MCBO implementation's GP-UCB rule. More generally, each quotient method preserves the acquisition rule of its base optimizer.

Thus an assumed graph H enters CBO through

$$H \mapsto (\mathbf{ES}(H), \{m_A^H\}_{A \in \mathbf{ES}(H)}) \mapsto \tau_{\mathcal{A}}(H), \quad (11)$$

where m_A^H is the prior-mean function computed under H and $\tau_{\mathcal{A}}(H)$ is the complete optimization trajectory: selected arms, intervention values, observations, and incumbents.

2.2 Quotients and cluster-union interventions

For a partition Π of $\mathbf{V}$, let $\chi : \mathbf{V} \rightarrow \Pi$ map each variable to its cluster. The quotient $\mathcal{G}^\Pi$ has a directed edge $\chi(u) \rightarrow \chi(v)$ whenever $u \rightarrow v$ is present in $\mathcal{G}$ and $\chi(u) \neq \chi(v)$; bidirected edges are mapped analogously. The partition is *valid* when the directed quotient is acyclic, and we write

$$Q_\Pi(\mathcal{G}) := (\Pi, \mathcal{G}^\Pi). \quad (12)$$

We assume that every cluster intersecting $\mathbf{X}$ is wholly manipulable; all other variables may remain singleton clusters. Let $\Pi_M := \{C \in \Pi : C \subseteq \mathbf{X}\}$ denote the manipulable clusters. The action family available at resolution Π is

$$\mathcal{U}(\Pi) = \left\{ \bigcup_{C \in S} C : \emptyset \neq S \subseteq \Pi_M \right\}. \quad (13)$$

QCBO applies the MIS rule to the quotient and expands each selected cluster arm to its constituent variables.

Let $\Pi_{\perp} := \{\{v\} : v \in \mathbf{V}\}$ denote the singleton partition. The best population value reachable under Π and its loss relative to the full fine action space are

$$V(\Pi) := \min_{A \in \mathcal{U}(\Pi)} \mu^*(A), \quad \Delta(\Pi) := V(\Pi) - V(\Pi_{\perp}). \quad (14)$$

We call $\Delta(\Pi)$ the *price of coarsening*. For minimization it is nonnegative. It measures the loss caused by the restricted action space. For example, if Π contains the manipulable cluster $\{X_1, X_2\}$, then the fine action family contains $\{X_1\}$, $\{X_2\}$, and $\{X_1, X_2\}$, whereas the coarse family contains only $\{X_1, X_2\}$.

For an incumbent intervention $\hat{a}_t$ after trial t , let $\mu(\hat{a}_t)$ be its expected outcome under the true SCM and define

$$y^* := V(\Pi_{\perp}), \quad r_t := \mu(\hat{a}_t) - y^*, \quad R_T := \sum_{t=1}^T r_t. \quad (15)$$

3 Robustness and the price of coarsening

3.1 Quotient invariance

For an ADMG $\mathcal{G}$ and valid partition Π , define the protected class

$$E_{\Pi}(\mathcal{G}) := \left\{ H : \begin{array}{l} H \text{ is an ADMG over } \mathbf{V}, \\ Q_{\Pi}(H) = Q_{\Pi}(\mathcal{G}) \end{array} \right\}. \quad (16)$$

This class contains within-cluster edits and cross-cluster edits that leave every quotient edge represented. It excludes edits that add or remove a quotient edge.

Let $\mathcal{D}$ denote the initial data, η the hyperparameters, and ω the optimizer's complete internal random stream (equivalently, its random seed together with all subsequent random draws). The quotient contract requires every graph-dependent operation to use a hypothesis graph only through Q_{Π} . We write $\mathcal{A}(H, \mathcal{D}, \omega, \eta)$ for a run of optimizer $\mathcal{A}$ under these inputs, and $\tau_{\mathcal{A}}(H; \mathcal{D}, \omega, \eta)$ for the resulting sequence of selected arms, intervention values, observations, and incumbents. In displays, let $\xi := (\mathcal{D}, \omega, \eta)$.

Proposition 1 (Exact quotient invariance). *Suppose*

$$\mathcal{A}(H, \mathcal{D}, \omega, \eta) = \bar{\mathcal{A}}(Q_{\Pi}(H), \mathcal{D}, \omega, \eta).$$

When two runs use the same true SCM, initial data, random stream, and hyperparameters,

$$Q_{\Pi}(H) = Q_{\Pi}(H') \implies \tau_{\mathcal{A}}(H; \xi) = \tau_{\mathcal{A}}(H'; \xi).$$

The two runs make the same first decision; because the true SCM is fixed, they receive the same cou-

pled observation; induction then gives identical subsequent decisions and observations. In Figure 1, deleting $X_1 \rightarrow Y$ is protected because $X_2 \rightarrow Y$ preserves the quotient edge, whereas deleting both parent edges removes $C_1 \rightarrow Y$ and is not protected.

3.2 Monotonicity under refinement

We write $\Pi' \preceq \Pi$ to mean that Π' *refines* Π : every cluster of Π' is contained in a cluster of Π . Refinement enlarges the set of representable interventions.

Proposition 2 (Monotonicity and price of coarsening). *For valid $\Pi' \preceq \Pi$,*

$$\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi'), \quad 0 \leq \Delta(\Pi') \leq \Delta(\Pi).$$

Moreover,

$$\Delta(\Pi) = 0 \iff \arg \min_{A \in \mathcal{U}(\Pi_{\perp})} \mu^*(A) \cap \mathcal{U}(\Pi) \neq \emptyset.$$

The result is purely about population reachability. Even when $\Delta(\Pi) = 0$, a partition can change intervention dimension, cost, arm count, and finite-budget behavior.

3.3 Sufficient conditions for lossless coarsening

For $A \subseteq \mathbf{X}$, define its cluster closure and the variables added by that closure as

$$\text{cl}_{\Pi}(A) := \bigcup \{C \in \Pi : C \cap A \neq \emptyset\}, \quad B := \text{cl}_{\Pi}(A) \setminus A.$$

For $a \in D(A)$, let B_a denote the joint natural response of B under $\text{do}(A = a)$ and define $D_B(a) := \{b : (a, b) \in D(\text{cl}_{\Pi}(A))\}$.

Proposition 3 (Sufficient conditions for zero price). *Assume a semi-Markovian SCM. If, for every $A \subseteq \mathbf{X}$ and $a \in D(A)$,*

$$\begin{aligned} \text{supp}(B_a) &\subseteq D_B(a), \\ B_a &\perp Y_{a,b} \quad \text{for every } b \in D_B(a), \end{aligned}$$

then $V(\Pi) = V(\Pi_{\perp})$ and $\Delta(\Pi) = 0$.

The first premise requires the intervention domain to contain the natural responses that are replaced when the remainder of a cluster is clamped. The second prevents the replacement from destroying a useful latent dependence with the target. We state graph-level conditions that imply these premises.

Definition 1 (Full intervention support). *Intervention domains are products, $D(W) = \prod_{w \in W} D(w)$ for every $W \subseteq \mathbf{X}$, and for every manipulable variable v and intervention a on $A \subseteq \mathbf{X} \setminus \{v\}$, $\text{supp}(v_a) \subseteq D(v)$.*

Definition 2 (Cluster-contained confounding). *Represent the semi-Markovian SCM by a nonparametric structural equation model with mutually independent exogenous variables and augmented graph $\mathcal{G}^{\text{aug}}$. Here nonparametric means that the structural mechanisms are not restricted to be linear, Gaussian, or members of a specified finite-dimensional family. Semi-Markovian SCMs allow latent confounding, represented by bidirected edges in the latent-projection ADMG, and strictly generalize Markovian SCMs, which exclude such confounding. This generality is needed for the confounded constructions studied in Sections 3.1 and 5. For $A \subseteq \mathbf{X}$ and $B = \text{cl}_{\Pi}(A) \setminus A$, let $\mathbf{U}_B(A)$ be the exogenous ancestors of B after $\text{do}(A)$ and let $\mathbf{U}_Y(A)$ be the exogenous ancestors of Y after $\text{do}(\text{cl}_{\Pi}(A))$. The partition has cluster-contained confounding when $\mathbf{U}_B(A) \cap \mathbf{U}_Y(A) = \emptyset$ for every $A \subseteq \mathbf{X}$.*

Corollary 1 (Graphical zero-price condition). *In a semi-Markovian SCM, full intervention support and cluster-contained confounding imply $V(\Pi) = V(\Pi_{\perp})$ and $\Delta(\Pi) = 0$.*

The condition is sufficient but not necessary. Because it constrains counterfactual dependence across interventions, it is not generally testable from a single observational or interventional distribution. It should therefore be read as a structural guarantee supplied by domain knowledge, rather than as a data-driven diagnostic.

3.4 Protected-class regret separation

Fix a true SCM $\mathcal{M}$ with graph $\mathcal{G}$. Outcomes are always generated by $\mathcal{M}$, while the optimizer receives a hypothesis graph H . Let $R_T(\mathcal{A}; H)$ denote the regret of a run using H at fixed $(\mathcal{D}, \omega, \eta)$. We denote worst-case regret over the protected class by “wc”:

$$R_T^{\text{wc}}(\mathcal{A}; \Pi, \mathcal{G}) := \sup_{H \in E_{\Pi}(\mathcal{G})} R_T(\mathcal{A}; H).$$

For a cluster-union incumbent, adding and subtracting the best value available at resolution Π gives

$$\mu(\hat{a}_t) - y^* = \underbrace{V(\Pi) - y^*}_{\text{coarsening gap}} + \underbrace{\mu(\hat{a}_t) - V(\Pi)}_{\text{within-space optimization error}}. \quad (17)$$

The first term is irreducible for a fixed action space and vanishes only when $\Delta(\Pi) = 0$. The second measures error relative to the best coarse action and can converge to zero.

Theorem 1 (Protected-class regret separation). *The following statements hold.*

- (i) *For every $\beta > 0$, there exist a semi-Markovian SCM, a valid partition with $\Delta(\Pi) = 0$, and a hypothesis $H \in E_{\Pi}(\mathcal{G})$ obtained by deleting one di-*

rected edge, such that every optimizer whose initial and sequential interventions lie in $\text{MIS}(H)$ and whose incumbent is selected among executed interventions satisfies

$$r_t \geq \beta \quad \text{for every } t, \quad R_T^{\text{wc}}(\mathcal{A}; \Pi, \mathcal{G}) \geq \beta T.$$

- (ii) *If an optimizer satisfies Proposition 1, executes only cluster-union interventions, and selects its incumbent among executed actions, then its trajectory is identical for every $H \in E_{\Pi}(\mathcal{G})$ and*

$$\begin{aligned} R_T^{\text{wc}}(\mathcal{A}; \Pi, \mathcal{G}) &= R_T(\mathcal{A}; \mathcal{G}) \\ &= T\Delta(\Pi) + \sum_{t=1}^T (\mu(\hat{a}_t) - V(\Pi)). \end{aligned}$$

Part (i) shows that a quotient-invisible edge error can force a fine-graph method to incur linear regret even when coarsening is lossless. The restrictions on both the initial design and incumbent selection are needed: otherwise an optimizer could evaluate or recommend an intervention excluded by $\text{MIS}(H)$. Part (ii) decomposes quotient regret into the unavoidable structural price $T\Delta(\Pi)$ and the remaining optimization error within the quotient action space.

3.5 Adaptive refinement

Proposition 4 (Linear regret floor). *For a fixed partition,*

$$r_t \geq \Delta(\Pi) \quad \text{for every } t, \quad R_T \geq T\Delta(\Pi).$$

Hierarchical Quotient Causal Bayesian Optimization (HQCBO) begins from a coarse partition and follows a declared refinement hierarchy. When progress stalls, it splits clusters participating in the incumbent, rejects splits whose quotient is cyclic, retains observations for arms that remain available, and initializes only newly exposed arms.

Definition 3 (Incumbent consistency). *The base optimizer is incumbent-consistent on the action space of a partition Π_J if continued optimization on that space satisfies $\mu(\hat{a}_t) \rightarrow V(\Pi_J)$ almost surely.*

Corollary 2 (Conditional refinement recovery). *If the refinement schedule visits a valid partition Π_J with $\Delta(\Pi_J) = 0$ and the base optimizer is incumbent-consistent on that action space, then*

$$r_t \rightarrow 0, \quad R_T/T \rightarrow 0.$$

The result is conditional on the declared hierarchy exposing a lossless partition and on the consistency of the base optimizer; it does not assert that the plateau trigger selects a useful split.

Corollary 3 (Adaptive refinement invariance). *Let $\tau_{\mathcal{H}}$ denote the HQCBO trajectory under a declared hierarchy $\mathcal{H}$. If the visited partitions are $\Pi_0, \dots, \Pi_J$, then*

$$[\forall j, \mathcal{G}^{\Pi_j} = (\mathcal{G}')^{\Pi_j}] \implies \tau_{\mathcal{H}}(\mathcal{G}) = \tau_{\mathcal{H}}(\mathcal{G}').$$

4 Quotient causal optimizers

Each quotient method replaces only the graph-dependent components of its base optimizer.

4.1 Arm-based QCBO

QCBO receives a manipulable partition and its quotient graph. For $S \subseteq \Pi_M$, let $U_{\Pi}(S) := \bigcup_{C \in S} C$. Its exploration set is

$$\mathbf{ES}^{\Pi} = \{U_{\Pi}(S) : \emptyset \neq S \subseteq \Pi_M, \quad S \subseteq \text{An}(Y) \text{ in } \mathcal{G}_S^{\Pi}\}. \quad (18)$$

For each arm, a complete identification procedure determines whether the observational distribution identifies its prior mean. When it does, QCBO evaluates the returned functional with GP plug-in estimates of the required observational conditionals. The implementation supports a sound cascade of backdoor adjustment, frontdoor adjustment, and g -computation; the complete identification check remains the gate, so g -computation is not used as a generic replacement for identification in an ADMG. Appendix B records the functionals and numerical implementation. The resulting two-tier prior is

$$m_A(a) = \begin{cases} \widehat{\mathbb{E}}[Y \mid \text{do}(A = a)] & \text{if identified in } \mathcal{G}^{\Pi}, \\ \bar{Y}_{\text{obs}} & \text{otherwise.} \end{cases} \quad (19)$$

Non-identifiable effects therefore receive an uninformative prior and are learned from interventions.

Proposition 5 (Identity recovery). *At $\Pi = \Pi_{\perp}$, the quotient graph equals the fine graph, the QCBO exploration set equals $\text{MIS}(\mathcal{G})$, and the prior construction coincides arm-by-arm with CBO.*

4.2 Dynamic and model-based variants

Quotient Dynamic Causal Bayesian Optimization (QDCBO) replaces each time-slice graph and its transitions by their quotients and uses joint cluster mechanisms within each slice [Aglietti et al., 2021]. Quotient Model-Based Causal Bayesian Optimization (QM-CBO) replaces node mechanisms by joint conditional cluster mechanisms

$$P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)}),$$

which are composed along the quotient graph [Sussex et al., 2023]. Joint rather than marginal mechanisms

are required to preserve dependence between variables within a cluster.

Corollary 4 (Portability of the quotient contract). *QDCBO satisfies Proposition 1 within each time slice, and QMCBO satisfies it for the composed model. At the singleton partition they recover DCBO and MCBO, respectively.*

The substituted objects are summarized below; the remaining acquisition and update logic is inherited from the base method.

Family	Quotient-level object
CBO $\rightarrow$ QCBO	intervention arms and causal priors
DCBO $\rightarrow$ QDCBO	time-slice cluster mechanisms
MCBO $\rightarrow$ QMCBO	joint cluster mechanisms

5 Experiments

5.1 Evaluation design and datasets

The evaluation separates four claims that would be confounded in a single aggregate benchmark: robustness to exploration-set corruption, robustness to causal-prior corruption, the action-space price of coarsening, and portability of the quotient contract. Table 1 summarizes this design. In every misspecification comparison, the true SCM, initial data, hyperparameters, and random stream are paired; only the graph supplied to the optimizer changes.

The controlled suite uses three SCMs with closed-form population effects: PARALLEL PARENT, FRONT DOOR, and MEDIATED CHAIN. Their fine graphs, partitions, and quotients appear in Appendix D. Each SCM provides one shared observational dataset; the optimizer receives its first $n_{\text{obs}} = 100$ observations. All displayed disturbance variables in this suite are mutually independent standard Gaussians; shared latent variables induce the stated bidirected edges. Interventional outcomes, including the initial design, are evaluated using the exact population expectation $\mathbb{E}[Y \mid \text{do}(A = a)]$, so these comparisons contain no target-simulation error. We use 30 paired seeds, three initial intervention points per arm, unit cost per intervened variable, and $T = 50$ trials, increased to 60 for MEDIATED CHAIN.

The portability suite follows each base method’s released protocol. The CBO family contains ToyGraph, CompleteGraph (“Synthetic”), and SimplifiedCoralGraph, with 10 seeds, $n_{\text{obs}} = 100$, ten initial points per arm, 40 trials, and minimization. The Coral simulator is the linear-SEM benchmark introduced by Aglietti et al. [2020]: its graph is based on a Bermuda reef ecosystem model and its mechanisms were fit to 50 field observations. The DCBO family uses station-

ary, independent, and nonstationary synthetic temporal SEMs, with 20 seeds, three time slices, ten trials per slice, and minimization. The MCBO family uses its ToyGraph and PSAGraph simulators with 20 seeds, $T = 100$, $\beta = 10$, zero observation noise, and maximization. PSAGraph is the model-based harness’s adaptation of the prostate-specific-antigen simulator used in the original CBO study.

We report five complementary quantities. The *best-so-far population value* evaluates the incumbent in the true SCM. For the controlled minimization suite, *cumulative incumbent regret* is Equation (15). *Paired trajectory deviation* compares the selected arms, intervention values, and incumbents of coupled correct- and wrong-graph runs. Exact invariance requires zero deviation, not merely equal final values. For the portability suite we report the final value as mean $\pm$ standard error and the paired quotient disadvantage with a paired- t 95% confidence interval. For minimization this difference is quotient minus base; for maximization it is base minus quotient, so positive values always indicate worse quotient performance.

5.2 Exploration-set corruption

The PARALLEL PARENT SCM is shown in Figure 1 and Appendix D. Its structural equations are

$$\begin{aligned} U &= 0.4\varepsilon_0, & X_1 &= U + 0.3\varepsilon_1, \\ X_2 &= U + 0.3\varepsilon_2, \\ Y &= (X_1 - 2)^2 + (X_2 + 2)^2 \\ &\quad + 0.1\varepsilon_Y. \end{aligned} \tag{20}$$

Under the correct graph, the fine MIS are $\{X_1\}$, $\{X_2\}$, and $\{X_1, X_2\}$, and the joint arm attains $y^* = 0$ at $(2, -2)$. Deleting $X_1 \rightarrow Y$ from the hypothesis graph removes every arm containing X_1 and leaves only $\{X_2\}$. Its exact population optimum is 4.25, so no amount of optimization under the wrong fine graph can close this gap.

After 50 trials, correct-graph CBO reaches 0.0016 ± 0.0002 , whereas misspecified CBO ends at 4.2500 ± 0.0000 (mean $\pm$ standard error). Their cumulative incumbent regrets are 4.21 ± 0.37 and 216.23 ± 1.12 , respectively. The latter value is close to the structural lower bound $4.25T = 212.5$; the excess is finite-budget optimization error within the misspecified action space.

For $\Pi = \{\{X_1, X_2\}, \{Y\}\}$, the same deletion is invisible because $X_2 \rightarrow Y$ preserves the quotient edge. The paired QCBO runs select the same arms and intervention values at every trial, and their incumbent trajectories have zero deviation. Figure 2(a) therefore separates a permanent fine-graph arm exclusion from exact

quotient invariance. A quotient-visible $X_1 \leftrightarrow Y$ perturbation serves as the negative control and changes the QCBO trajectory, with maximum paired incumbent deviation 21.45.

5.3 Prior corruption without arm loss

The FRONTDOOR graph in Appendix D contains $X_1 \rightarrow M \rightarrow Y$ and latent $X_1 \leftrightarrow Y$. Its narrow-well SCM is

$$\begin{aligned} U &= 0.4\varepsilon_0, & X_1 &= U + 0.3\varepsilon_1, \\ M &= 2X_1 + 0.2\varepsilon_M, \\ Y &= 2 \left[1 - \exp \left\{ -\frac{(M-1)^2}{2(0.3)^2} \right\} \right] \\ &\quad + U + 0.1\varepsilon_Y. \end{aligned} \tag{21}$$

Its population objective is minimized by $\text{do}(M = 1)$. Under the correct graph, $\text{do}(M)$ is identified by adjustment for X_1 and $\text{do}(X_1)$ by the frontdoor functional. Adding a spurious $X_1 \leftrightarrow M$ leaves the fine MIS unchanged but makes both prior means non-identifiable. This design therefore changes the prior channel while holding the available arms fixed.

Fine CBO loses its observational head start and accumulates additional regret while searching for the narrow optimum, but it can recover from interventional observations because the optimal arm remains available. The paired increase in cumulative regret is 5.91 ± 1.04 at $T = 50$; the final values are 0.0003 ± 0.0003 under the correct graph and 0.0035 ± 0.0022 under the corrupted graph. This is a transient finite-budget effect rather than an action-space exclusion. The perturbation is internal to $\{X_1, M\}$ and disappears in the quotient, so the paired QCBO decisions and incumbent trajectories are identical. The right panel of Figure 2 contrasts this transient fine-graph effect with exact quotient invariance.

5.4 Action-space price and refinement

The MEDIATED CHAIN graph in Appendix D has $X_1 \rightarrow X_2 \rightarrow Y$, with $X_2 = 2X_1 + \varepsilon_2$ and $Y = (X_2 - 4)^2 + \varepsilon_Y$. The fine arm $\text{do}(X_1 = 2)$ leaves X_2 free to respond naturally around 4 and attains $y^* = \text{Var}(\varepsilon_2) = 0.04$. In contrast, fixed QCBO has only the whole-cluster arm $\{X_1, X_2\}$. This arm must clamp X_2 inside its intervention domain $[-2, 2]$; once X_2 is clamped, X_1 has no remaining effect on Y . The best allowed choice is therefore $X_2 = 2$, which yields

$$V(\Pi) = 4, \quad y^* = 0.04, \quad \Delta(\Pi) = 3.96.$$

Consequently, every fixed-QCBO incumbent has instantaneous regret at least 3.96, and Proposition 4

Table 1: Evaluation map. Arrows give the objective direction.

Study	Data	Perturbation	Comparison	Objective	Primary metric	Purpose
A	PARALLEL PARENT	delete $X_1 \rightarrow Y$	CBO versus QCBO	$\downarrow$	regret; deviation	exploration-set robustness
B	FRONT DOOR	add $X_1 \leftrightarrow M$	CBO versus QCBO	$\downarrow$	regret; deviation	prior robustness
C	MEDIATED CHAIN	refine $\{X_1, X_2\}$	CBO, QCBO, HQCBO	$\downarrow$	value; regret	price and recovery
D	eight family tasks	quotient graph view	base versus quotient	$\downarrow / \uparrow$	final value; paired CI	portability

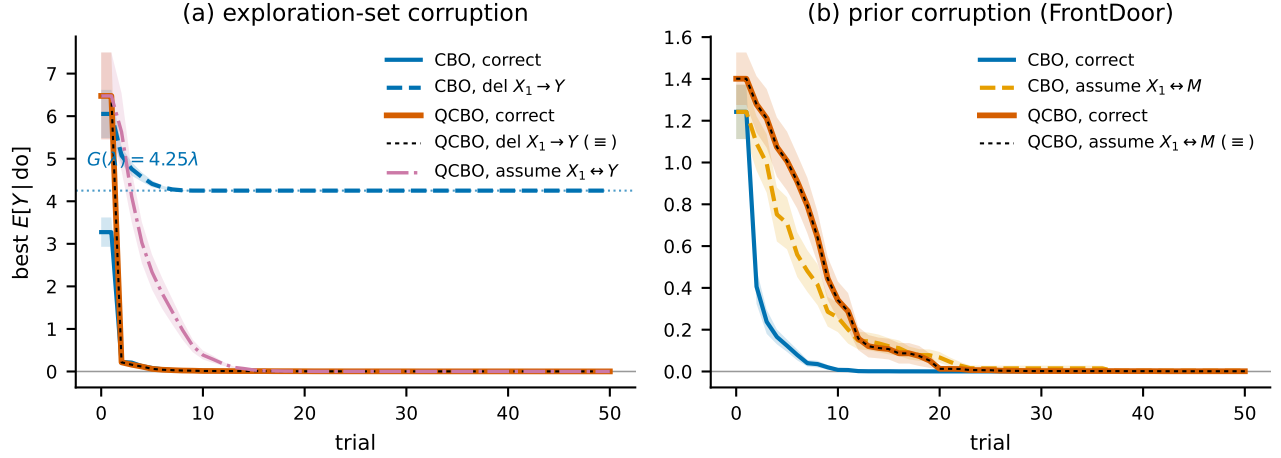


Figure 2: Correct and misspecified trajectories, mean $\pm$ standard error over 30 seeds. Left: a wrong fine graph removes the optimal arm, while QCBO is unchanged for a quotient-invisible edit. Right: prior corruption slows fine CBO transiently, while QCBO remains exactly invariant.

gives the permanent lower bound $R_T \geq 3.96T$. Fine CBO does not incur this floor because its MIS retains $\{X_1\}$. HQCBO removes it only after the declared hierarchy splits $\{X_1, X_2\}$ and exposes the singleton arm. Figure 3 shows the resulting best-value and cumulative-regret trajectories. CBO, fixed QCBO, and HQCBO finish at 0.0400 ± 0.0000 , 4.0000 ± 0.0000 , and 0.0400 ± 0.0000 , with cumulative regrets 5.46 ± 1.61 , 243.23 ± 0.90 , and 36.97 ± 1.18 , respectively. HQCBO refines at trial 7.4 ± 0.1 ; all 30 declared splits are accepted. The fixed-QCBO result exceeds the theoretical floor $60 \times 3.96 = 237.6$ only by its within-space optimization error. The experiment demonstrates recovery for this declared split; it does not imply that the plateau rule always selects a useful refinement.

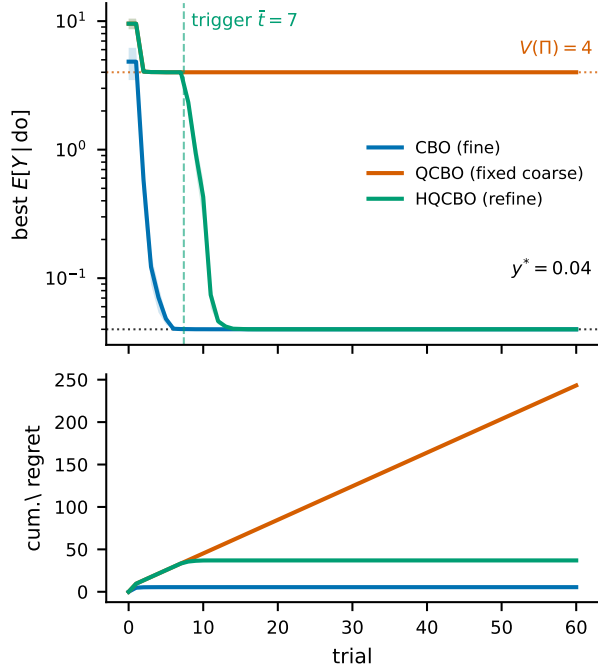
5.5 Portability across optimizer families

Figure 4 compares each quotient construction with its base optimizer on the eight tasks described above; all corresponding DAGs and C-DAGs are in Appendix D. The structural prediction is exact: QDCBO is invariant in all 60 paired dynamic runs and QMCBO in all 40 paired model-based runs, whereas their base methods change under the same graph-view perturbations. This confirms that the quotient contract is independent of the family-specific acquisition and surrogate construction.

Finite-budget performance is deliberately reported separately from invariance. Table 2 gives the final population value and paired quotient disadvantage for each task. For $\downarrow$ rows the paired difference is quotient minus base; for $\uparrow$ rows it is base minus quotient. Positive values therefore always mean that the quotient ended worse. These are finite-budget effects incorporating optimization and initialization, not estimates of the theoretical action-space quantity $\Delta(\Pi)$.

Table 2: Final population values and paired quotient disadvantage. Each task label carries its optimization direction. Positive paired effects mean worse quotient performance; intervals are paired- t 95% CIs.

Experiment	Base final	Quotient final	Paired quotient disadvantage [95% CI]
<i>CBO vs QCBO (min; $n=10$ paired seeds)</i>			
Toy $\downarrow$	-2.16 ± 0.00	-2.13 ± 0.03	$+0.03$ $[-0.04, +0.11]$
Synthetic $\downarrow$	-3.14 ± 0.19	-1.25 ± 0.07	$+1.89$ $[+1.40, +2.39]$
Coral $\downarrow$	36.08 ± 0.02	36.42 ± 0.01	$+0.34$ $[+0.30, +0.37]$
<i>DCBO vs QDCBO (min; $n=20$ paired seeds)</i>			
stat (T=3) $\downarrow$	-6.14 ± 0.05	-6.43 ± 0.02	-0.29 $[-0.41, -0.17]$
ind (T=3) $\downarrow$	-3.12 ± 0.06	-5.57 ± 0.15	-2.45 $[-2.72, -2.19]$
nonstat (T=3) $\downarrow$	8.03 ± 0.90	6.33 ± 0.29	-1.69 $[-3.57, +0.18]$
<i>MCBO vs QMCBO (max; $n=20$ paired seeds)</i>			
ToyGraph $\uparrow$	1.39 ± 0.13	2.16 ± 0.00	-0.77 $[-1.04, -0.50]$
PSAGraph $\uparrow$	-5.15 ± 0.00	-5.15 ± 0.00	$+0.0001$ $[-0.0004, +0.0006]$


 Figure 3: MEDIATEDCHAIN over 30 seeds. Top: mean $\pm$ standard error best-so-far population value and mean refinement time. Bottom: mean cumulative incumbent regret, excluding the shared initial incumbent. Fixed QCBO cannot execute $\text{do}(X_1)$; refinement exposes it.

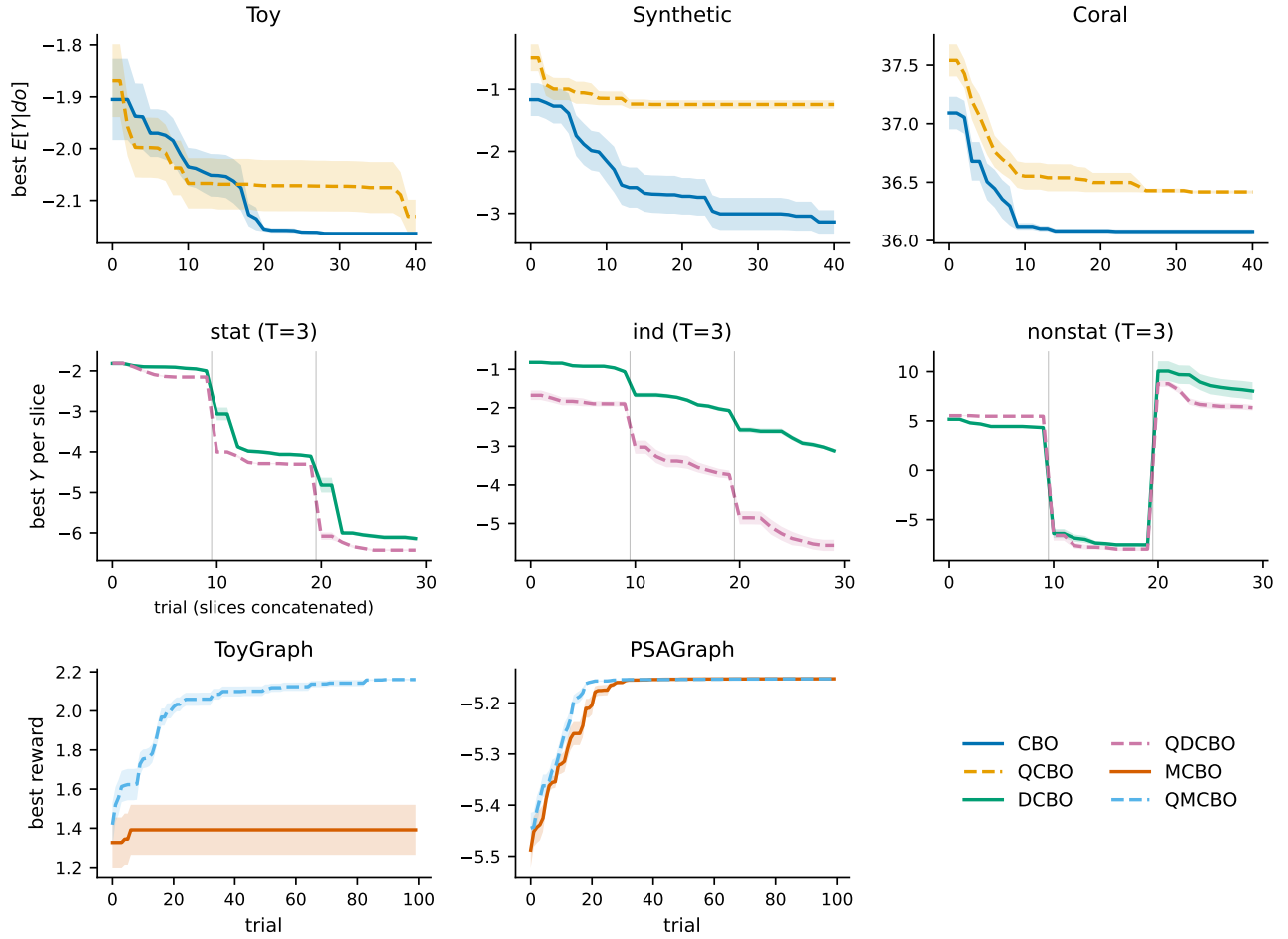


Figure 4: Quotient methods and their bases under the released family protocols, mean $\pm$ standard error. Top: CBO minimization. Middle: dynamic minimization over three time slices. Bottom: MCBO maximization.

6 Related work

Graph-agnostic and causal entropy optimization methods learn over candidate graphs online [Mukherjee et al., 2024, Branchini et al., 2023], while causal-bandit guarantees price learning an unknown graph within the horizon [Lu et al., 2021]. Our setting instead studies commitment to a fixed coarse input. These approaches are complementary: a graph learner could operate over, or trigger refinement of, a quotient.

The construction builds on cluster-DAG identification [Anand et al., 2023] and is related to causal abstraction and macro-level structure learning [Beckers and Halpern, 2019, Madaleno et al., 2026]. We coarsen manipulable inputs and explicitly characterize the actions removed by that choice.

7 Limitations

Exact invariance requires the correct quotient; quotient-visible errors and incorrect partitions remain harmful. The zero-price condition is sufficient and structural, and its cross-world consequence is not directly testable from single-world data. Refinement is heuristic, with recovery conditional on visiting a lossless partition and on consistency of the base optimizer. Different partitions also induce different arm counts, initialization costs, and intervention costs. Finally, QMCBO inherits the causal-sufficiency assumption of its model-based factorization.

8 Conclusion

Coarsening makes precisely those graph distinctions that disappear in the quotient irrelevant to optimization. Within the resulting protected class, the complete trajectory is invariant. The price is the optimal value of fine interventions excluded by the partition, which is monotone under refinement and zero under the stated structural conditions. Theoretical and empirical results show that quotienting can prevent catastrophic graph errors, while hierarchical refinement provides a conditional route back to the fine action space.

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A Proofs and additional technical results

A.1 Minimal intervention sets and quotient values

Lemma 1 (MIS reduction). *Let G be an ADMG with manipulable vertices M and target Y . For $\emptyset \neq W \subseteq M$, let $W' = W \cap \text{An}(Y)$ in $G_{\bar{W}}$. Then*

- (i) $P(Y \mid \text{do}(W = w)) = P(Y \mid \text{do}(W' = w|_{W'}))$;
- (ii) W' is a MIS, or is empty if no member reaches Y ; and
- (iii) restricting the action search to $\text{MIS}(G)$ does not change the minimum population value.

Proof. In $G_{\bar{W}}$, variables in $W \setminus W'$ have no directed path to Y , so dropping them from the intervention does not change the distribution of Y . Every directed path from a member of W' to Y avoids all intervened vertices and remains after the edges into $W \setminus W'$ are restored; hence W' satisfies the MIS condition. Applying this reduction to every candidate intervention proves the equality of minima. $\square$

Consequently, the best value over the quotient MIS expanded to fine variables equals $V(\Pi)$ in Equation (14).

A.2 Proofs of the main structural results

Proof of Proposition 1. Write the run as decisions $d_1, d_2, \dots$ and datasets $\mathcal{D}_0, \mathcal{D}_1, \dots$, with

$$d_t = f_t(Q_\Pi(H), \mathcal{D}_{t-1}, \omega_t, \eta).$$

Equal quotients, initial data, random streams, and hyperparameters imply equal first decisions. The fixed true SCM produces the same coupled observation, so the updated datasets also agree. Induction gives equality of every decision, observation, incumbent, and stopping event. $\square$

Proof of Proposition 2. Every union of clusters in Π is a union of clusters in a refinement Π' , and every such union is available at the singleton partition. The three population minima are therefore taken over nested action families: $V(\Pi_\perp) \leq V(\Pi') \leq V(\Pi)$. Equality between the first and last terms holds exactly when a global fine optimum belongs to $\mathcal{U}(\Pi)$. $\square$

Proof of Proposition 3. Fix A and a . Consistency and cross-world independence give

$$\begin{aligned} \mathbb{E}[Y_a] &= \int \mathbb{E}[Y_{a,b} \mid B_a = b], dP(B_a = b) \\ &= \int \mathbb{E}[Y_{a,b}], dP(B_a = b) \\ &\geq \inf_{b \in D_B(a)} \mathbb{E}[Y_{a,b}], \end{aligned}$$

where support inclusion justifies the final inequality. Thus the whole cluster closure can match or improve every fine intervention: $\mu^*(\text{cl}_\Pi(A)) \leq \mu^*(A)$. Taking minima and using $\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi_\perp)$ gives equality. $\square$

Lemma 2 (Cluster containment implies cross-world independence). *Under the model in Definition 2, cluster-contained confounding implies $B_a \perp Y_{a,b}$ for every A , a , and $b \in D_B(a)$.*

Proof. After $\text{do}(A = a)$, recursive substitution expresses B_a as a measurable function of $\mathbf{U}_B(A)$. After clamping the entire cluster closure, it expresses $Y_{a,b}$ as a measurable function of $\mathbf{U}_Y(A)$. The two exogenous ancestor sets are disjoint by Definition 2 and the exogenous variables are mutually independent, so the two functions are independent. Full intervention support separately implies $\text{supp}(B_a) \subseteq D_B(a)$, proving Corollary 1. $\square$

Proof of Theorem 1. For part (i), use PARALLEL PARENT with manipulable variables $\{X_1, X_2\}$, domains $[-3, 3]^2$, edges $X_1 \rightarrow Y$ and $X_2 \rightarrow Y$, latent $X_1 \leftrightarrow X_2$, and partition $\{\{X_1, X_2\}, \{Y\}\}$. Choose the objective curvature so that the best $\{X_2\}$ -only intervention has value β , while the joint arm attains $y^* = 0$. Deleting $X_1 \rightarrow Y$ preserves

the quotient through $X_2 \rightarrow Y$ but leaves $\text{MIS}(H) = \{\{X_2\}\}$. Every executed intervention and every eligible incumbent therefore has regret at least β , giving $R_T \geq \beta T$. The whole-cluster arm remains globally optimal, so $\Delta(\Pi) = 0$.

For part (ii), Proposition 1 makes the trajectory and regret constant over $E_\Pi(\mathcal{G})$. Adding and subtracting $V(\Pi)$ in each regret term gives

$$r_t = \Delta(\Pi) + \mu(\hat{a}_t) - V(\Pi).$$

The second term is nonnegative because every incumbent is an executed cluster-union action. Summing over t yields the stated decomposition. $\square$

Proof of Proposition 4. Every incumbent belongs to the fixed action family, hence $\mu(\hat{a}_t) \geq V(\Pi)$. Subtracting $y^* = V(\Pi_\perp)$ gives $r_t \geq \Delta(\Pi)$; summation gives the cumulative bound. $\square$

Proof of Corollary 2. At a visited lossless partition, incumbent consistency gives $\mu(\hat{a}_t) \rightarrow V(\Pi_J) = y^*$, hence $r_t \rightarrow 0$ almost surely. Cesàro averaging then gives $R_T/T \rightarrow 0$. $\square$

Proof of Corollary 3. Proceed by induction over refinement phases. Within phase j , all graph-dependent behavior factors through $(\Pi_j, \mathcal{G}^{\Pi_j})$. Equal quotients and equal trajectory prefixes therefore produce the same phase trajectory, plateau decision, and next split. Repeating this argument through Π_J proves equality of the adaptive trajectories. $\square$

A.3 Necessity, refinement validity, and model-based soundness

Proposition 6 (Necessity of the quotient contract). *If, for every ADMG $\mathcal{G}$, the trajectory map is constant on $E_\Pi(\mathcal{G})$ at fixed $(\mathcal{D}, \omega, \eta)$, then the trajectory map factors through Q_Π .*

Proof. Constancy on every fiber of Q_Π makes $\bar{\tau}(q) := \tau(H)$ well defined for any H satisfying $Q_\Pi(H) = q$. Hence $\tau = \bar{\tau} \circ Q_\Pi$. $\square$

Proposition 7 (Refinement can invalidate a quotient). *There exist a valid partition Π and a refinement $\Pi' \preceq \Pi$ for which $\mathcal{G}^\Pi$ is acyclic but $\mathcal{G}^{\Pi'}$ contains a directed cycle. Validity must therefore be checked after every split.*

Proof. Let the fine graph contain $a_1 \rightarrow b_1$ and $b_2 \rightarrow a_2$. The single cluster $\{a_1, a_2, b_1, b_2\}$ has an acyclic one-node quotient. Refining it into $\{a_1, a_2\}$ and $\{b_1, b_2\}$ exposes both quotient directions and produces a two-cycle. $\square$

Proposition 8 (Soundness of joint cluster mechanisms). *Under causal sufficiency and a valid partition, every intervention on a union of clusters S satisfies*

$$P(\mathbf{V} \mid \text{do}(\mathbf{x}_S)) = \prod_{C \notin S} P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})|_{\mathbf{x}_S = \mathbf{x}_S}.$$

Proof. Apply the fine truncated factorization and group its factors by clusters. Within each cluster, the chain rule combines the member-wise factors into the joint conditional mechanism. Validity orders these mechanisms along the quotient DAG. The joint conditional is essential: replacing it by the product of member-wise marginals would discard within-cluster dependence. $\square$

Proposition 9 (Sub-cluster effects are not identified by cluster mechanisms). *There exist two causally sufficient SCMs with the same valid quotient, joint cluster mechanisms, and whole-cluster interventional distributions, but different values of $P(Y \mid \text{do}(A = a))$ for a strict subset A of a cluster.*

Proof. Let $C = \{X_2, X_3\}$ be a root cluster and $Y = X_2 + \varepsilon_Y$. In one SCM take $X_2 = \varepsilon_2$ and $X_3 = aX_2 + \varepsilon_3$; in another, choose the reverse linear-Gaussian factorization of the same joint distribution. The two models agree on the joint law of C and on every whole-cluster intervention. Under $\text{do}(X_3 = x)$, however, X_2 remains mean zero in the first model and changes linearly with x in the second. Thus quotient-respecting inputs do not identify the sub-cluster effect. $\square$

B Identification and numerical estimation

The complete identification check is performed on the ADMG supplied to the optimizer. When the query is identified, the returned functional is instantiated by a sound library of GP plug-in estimators. For a valid backdoor set Z , the estimator is Equation (6). For a frontdoor mediator M between treatment X and target Y , it uses

$$\hat{p}(y \mid \text{do}(x)) = \int \hat{p}(m \mid x) \left\{ \int \hat{p}(y \mid m, x') d\hat{P}(x') \right\} dm. \quad (22)$$

Under causal sufficiency it instead evaluates the truncated factorization in Equation (7). Required conditional means and densities are fitted from the observational sample with GP regressions and integrated numerically over the intervention domain. If complete identification fails, or if the identified functional lies outside this sound implementation library, the arm receives the common uninformative prior $m_A(a) = \bar{Y}_{\text{obs}}$. Interventional observations subsequently update the arm’s GP in the usual way.

This distinction is important: complete identification determines whether a causal effect is a functional of the observational law, whereas backdoor, frontdoor, and g -computation describe the particular functionals currently implemented. The latter list is deliberately not presented as complete for arbitrary ADMGs.

C Experimental details

Controlled suite. All methods receive the same sampled observational dataset and exact population intervention evaluator. Initial intervention values are sampled uniformly from the declared domains and their outcomes are computed by the same closed-form evaluator used during sequential optimization. Each run logs the selected arm and intervention values in addition to the incumbent. For protected perturbations, arm sequences and intervention values are compared exactly and incumbent values to 10^{-8} , allowing for numerical GP optimization. The complete perturbation taxonomy is reported in Table 3.

Table 3: Controlled misspecifications and measured paired trajectory deviations. Rows whose quotient is unchanged are predicted invariant by Proposition 1.

Cond.	Edit (assumed graph)	Quotient changed	Fine MIS changed	Fine prior changed	sup $ \Delta $
A1 (PP)	del $X_1 \rightarrow Y$ (quotient-redundant)	×	✓	—	0
A2 (PP)	add $X_1 \rightarrow X_2$ (intra-cluster bow)	×	×	✓	0
A3 (PP)	assume $X_1 \leftrightarrow Y$ (quotient-visible)	✓	×	✓	21.5
B1 (FD)	assume $X_1 \leftrightarrow M$ (intra-cluster conf.)	×	×	✓	0

Family suite. The CBO/QCBO comparison uses a common arm-based implementation with cost-normalized EI. DCBO/QDCBO and MCBO/QMCBO retain the released base implementations, including their acquisition and update schedules. Thus the portability experiment tests whether quotient invariance survives these family-specific choices rather than forcing every method through one optimizer. All plots show means and standard errors over paired seeds. The paired intervals in Table 2 use the seedwise differences defined in Section 5.1.

D Dataset graphs and their quotients

For every distinct dataset topology, the following figures show the fine graph with its coarse partition and the quotient graph consumed by the optimizer. Manipulable variables are orange, targets are blue, dash-dotted edges denote latent confounding, and dashed boxes indicate manipulable clusters. Graph-isomorphic benchmark pairs are shown once under their shared topology.



Figure 5: **ParallelParent (MinimalBench, Exp. A)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The intra-cluster $X_1 \leftrightarrow X_2$ disappears in the quotient; deleting $X_1 \rightarrow Y$ leaves $C_1 \rightarrow Y$ alive through $X_2 \rightarrow Y$ (quotient-redundant).

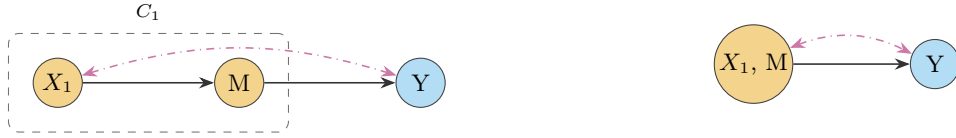


Figure 6: **FrontDoor (MinimalBench, Exp. B) / ToyGraph (CBO family)**. These datasets share the same fine and quotient topology under $X_1 \mapsto X$ and $M \mapsto Z$. The quotient is a bow ($C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$): the cluster arm uses the uninformative prior tier. In FRONTDOOR, the assumed intra-cluster $X_1 \leftrightarrow M$ therefore cannot change it.

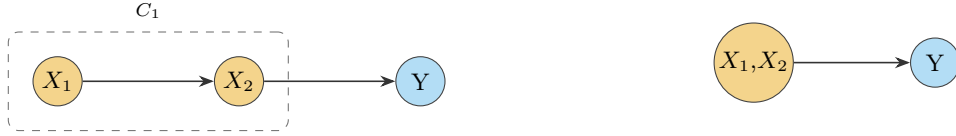


Figure 7: **MediatedChain (MinimalBench, Exp. C) / ToyGraph-M (QMCBO family)**. These datasets share the same chain-and-partition topology under $X_1 \mapsto X_0$ and $X_2 \mapsto X_1$. For MEDIATEDCHAIN, the policy box restricts $D(X_2)$ to $[-2, 2]$; the fine optimum $\text{do}(X_1=2)$ drives X_2 to 4, outside the box.

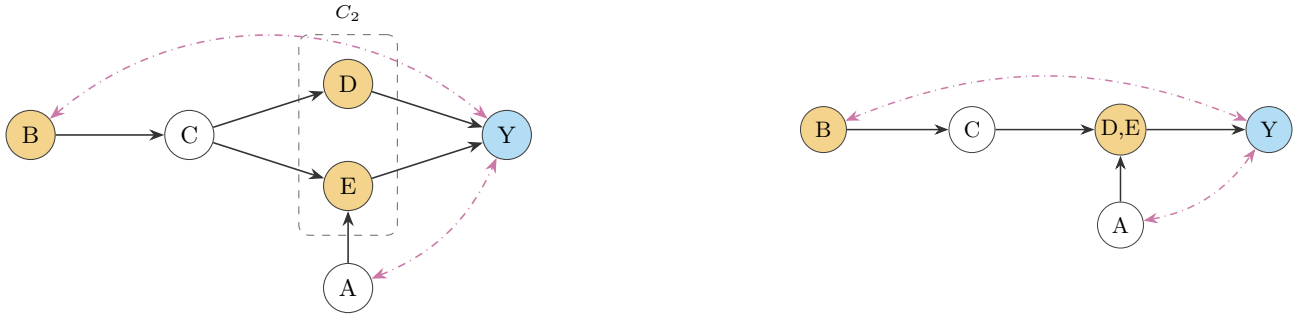


Figure 8: **CompleteGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

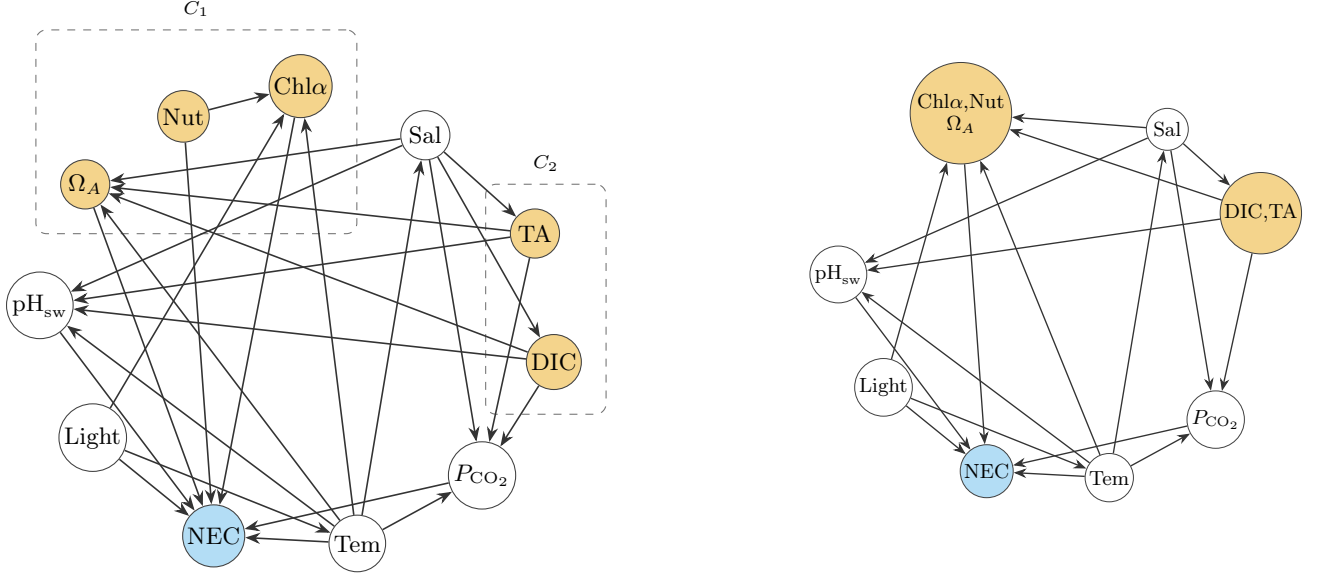


Figure 9: **SimplifiedCoralGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

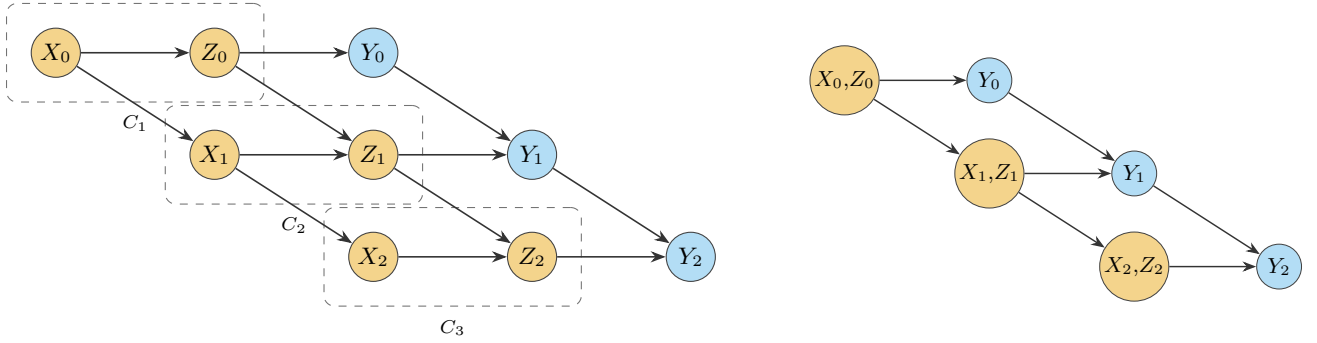


Figure 10: **DCBO stat / nonstat (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. Slices $t = 0, 1, 2$; the nonstat setup shares this topology with a change point in the SEM.

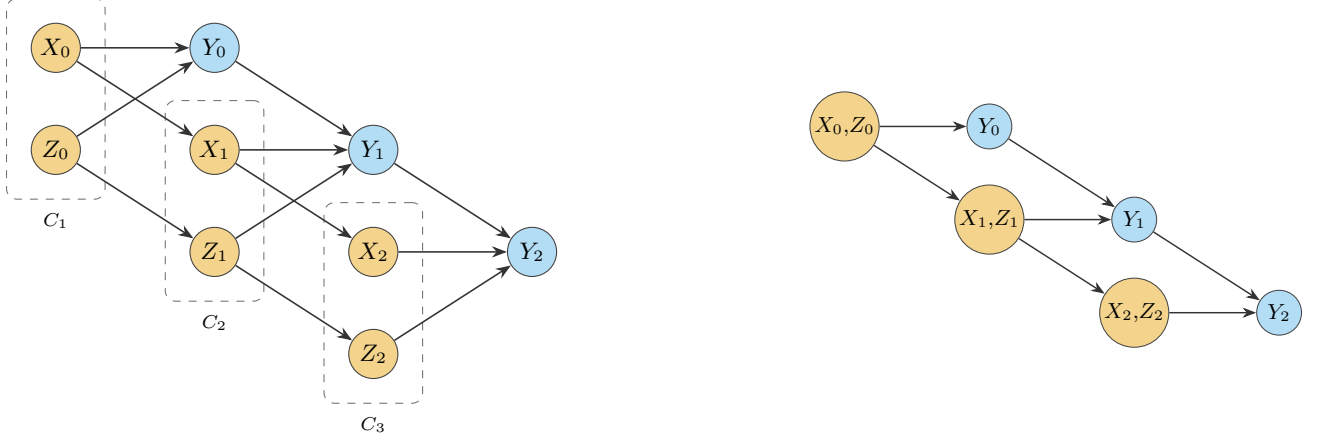


Figure 11: **DCBO ind (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

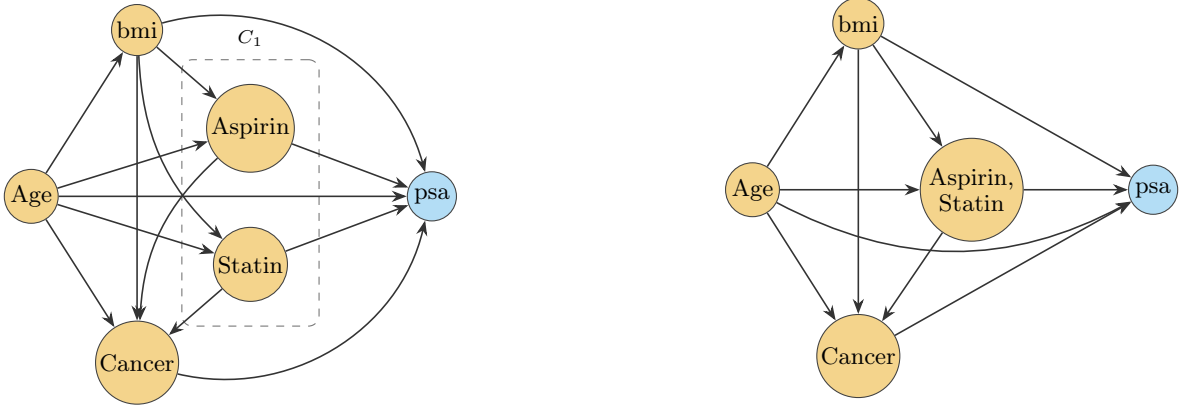


Figure 12: **PSAGraph (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.